Proof-check packet: the K-group lower bound

Paper 1, "Auditable Memory and Financial Identification Limits", Theorem 6.1 (`thm:kcohort`)

Prepared for an independent reader · 8 November 2026

What we are asking

This is the headline result of the paper: separating K trader groups whose memories are $\Delta\eta$ apart takes of order $(\Delta\eta)^{-(4K-2)}$ experiments, and the feedback loop does not change that order. We ask you to do two things, which should take two to three hours.

1. **Check the proof** in Section 2. It is split into numbered claims. For each, record *checked*, *objection*, *corrected* or *open* in the table on the last page.
2. **Check that the machine-checked statement says what the paper says** (Section 3). Lean has verified a version of the bound. The question we cannot answer ourselves is whether its hypotheses and conclusion match the paper's claim, and where they differ, whether the difference matters.

Why an outside reader. The paper was written with substantial AI assistance, and its internal reviews were AI-generated. Those do not count as independent review. The co-authors are not independent either. Your check would be the first independent one. You will be acknowledged by name only if you agree.

1 Setting

Market. A single asset has pricing error x_t. A dealer moves the price by $\kappa > 0$ per unit of unexpected flow, and the error persists with coefficient $a \geq 0$. K groups of feedback traders each hold a position that tracks past pricing errors with exponential memory:

$$u_{i,t} = r_i u_{i,t-1} + w_i x_{t-1}, \qquad 0 < r_i < 1,\ w_i > 0.$$

Their combined effect on the price enters only through the *kernel* $k_m = \sum_i w_i r_i^m$ (the power moments of the merged measure $\sum_i w_i \delta_{r_i}$).

Reset experiment. From rest, an observer injects a unit impulse and records the price response

$$h_0 = \kappa, \qquad h_{t+1} = a\, h_t + \kappa \sum_{j=0}^{t} k_j\, h_{t-j},$$

observed with i.i.d. $N(0, \sigma^2)$ noise on each of T taps. N independent experiments are run.

Spacing. The rates lie in $r_c + \Delta\eta[-1, 1]$, written $r_i = r_c + \Delta\eta\, s_i$ with $|s_i| \leq 1$.

Theorem (K-cohort two-point lower bound; paper's statement)**.** *Fix $K \geq 2$, $r_c \in (0,1)$, $W > 0$ and $0 < \Delta\eta < \min(r_c, 1 - r_c)$, and assume*

$$a + \kappa W/(1 - r_c - \Delta\eta) < 1.$$

Each reset experiment observes the noisy taps $h_0, \ldots, h_{T-1}$ with $T \geq 2K + 1$ ($T = \infty$ allowed), so that the first differing tap h_{2K} is observed. Then there exist two K-group economies with the following properties:

- *their rates lie in $r_c + \Delta\eta[-1, 1]$ and their weights are positive;*
- *the moments of their merged measures of order $0, \ldots, 2K - 2$ coincide, and the moments of order $2K - 1$ differ by a constant times $(\Delta\eta)^{2K-1}$;*
- *their impulse responses differ by $\|h^A - h^B\|_2 \asymp (\Delta\eta)^{2K-1}$.*

Recovering the weights from N independent noisy reset experiments, with worst-case error probability below $1/4$, therefore requires $N \geq c\,\sigma^2(\Delta\eta)^{-(4K-2)}$. The same holds for the loss $W_1(\hat{M}, M)$ between estimated and true merged measures, at accuracy $c_K W \Delta\eta$.

2 Proof, step by step

Step 1: the pair. Let (s^A, ω^A) and (s^B, ω^B) be the K-point Gauss and Gauss–Radau rules for the uniform weight on $[-1, 1]$, normalized to unit mass. Put $r_i^X = r_c + \Delta\eta\, s_i^X$ and weights $W\omega_i^X$. Write $\mu_j^X = W \sum_i \omega_i^X (s_i^X)^j$.

Claim 1. *Both rules have positive weights and nodes in $[-1, 1]$. Both integrate polynomials of degree $\leq 2K - 2$ exactly, so $\mu_j^A = \mu_j^B$ for $j \leq 2K - 2$. Their moments of degree $2K - 1$ differ, $\mu_{2K-1}^A - \mu_{2K-1}^B = W g_K$ with $g_K \neq 0$. Also $|\mu_j^X| \leq W$ for all j.*

Notes. Gauss is exact to degree $2K - 1$ and Gauss–Radau to degree $2K - 2$. So their difference at degree $2K - 1$ is minus the Radau error functional applied to x^{2K-1}, which is nonzero. Numerically, $|g_K| = 0.222, 0.0533, 0.0131$ for $K = 2, 3, 4$. Analytically, for every K: with Radau nodes $-1, s_2, \ldots, s_K$, the polynomial $P(s) = (s+1)\prod_{j \geq 2}(s - s_j)^2$ is monic of degree $2K - 1$ and vanishes at every Radau node. It is nonnegative on $[-1, 1]$, so Gauss (exact to degree $2K - 1$) gives it $\int P\, d\mu > 0$, while Radau gives it 0. The rules agree on lower degrees, so $g_K \neq 0$. (This argument was suggested by an AI-assisted review.)

Step 2: kernel gap.

Claim 2. $k_m^X = \sum_{j=0}^m \binom{m}{j} r_c^{m-j} (\Delta\eta)^j \mu_j^X$.

Claim 3. *For $j \geq n$: $\binom{m}{j} \leq \binom{m}{n}\binom{m-n}{j-n}$. Hence, with $n = 2K - 1$,*

$$|\Delta k_m| = |k_m^A - k_m^B| \leq 2W \binom{m}{n} (\Delta\eta)^n (r_c + \Delta\eta)^{m-n} \quad (m \geq n), \qquad \Delta k_m = 0 \quad (m < n).$$

Hint. $\binom{m}{j}\binom{j}{n} = \binom{m}{n}\binom{m-n}{j-n}$ and $\binom{j}{n} \geq 1$.

Claim 4. $\sum_{m \geq 0} |\Delta k_m| \leq 2W(\Delta\eta)^{2K-1}/(1 - r_c - \Delta\eta)^{2K}$, *using* $\sum_{m \geq n} \binom{m}{n} x^{m-n} = (1 - x)^{-(n+1)}$ *for $0 \leq x < 1$.*

Step 3: response gap, upper order. Let $\varsigma = a + \kappa W/(1 - r_c - \Delta\eta) < 1$.

Claim 5. $\sum_m |k^X_m| \le W/(1 - r_c - \Delta\eta)$. *Every partial sum* $\sum_{t<n} |h^B_t| \le \kappa/(1 - \varsigma)$.

Hint. Sum the recursion over $t < n$. Then $\sum_{t\le n} |h_t| \le \kappa + \varsigma \sum_{t<n} |h_t|$, and induct.

Claim 6. $\Delta h = h^A - h^B$ *satisfies* $\Delta h_0 = 0$ *and*

$$\Delta h_{t+1} = a\,\Delta h_t + \kappa \sum_{j\le t} \big(k^A_j\,\Delta h_{t-j} + \Delta k_j\,h^B_{t-j}\big).$$

Hence every partial sum of $|\Delta h|$ *is at most* $\kappa\|\Delta k\|_1\|h^B\|_1/(1 - \varsigma)$.

Claim 7. *For every horizon,* $\|h^A - h^B\|_2 \le \|h^A - h^B\|_1 \le C_U(\Delta\eta)^{2K-1}$, *where*

$$C_U = \frac{2\kappa^2 W}{(1 - r_c - \Delta\eta)^{2K}(1 - \varsigma)^2}.$$

Step 4: response gap, lower order.

Claim 8. $\Delta h_t = 0$ *for* $t \le 2K - 1$, *and* $\Delta h_{2K} = \kappa^2 W g_K(\Delta\eta)^{2K-1}$. *So when the taps* $h_0, \dots, h_{T-1}$ *are observed with* $T \ge 2K + 1$, $\|h^A - h^B\|_2 \ge \kappa^2 W|g_K|(\Delta\eta)^{2K-1}$. *With* $T \le 2K$ *the observed responses coincide. (An earlier version of this packet said* $T \ge 2K$, *which is off by one.)*

Hint. Δh_{t+1} involves only Δk_j with $j \le t$, and $\Delta k_{2K-1} = (\Delta\eta)^{2K-1} W g_K$ by Claim 2.

Step 5: from the response gap to the sample bound.

Claim 9. *The laws of* N *independent experiments with* T *taps have Kullback–Leibler divergence* $N \sum_{t<T}(h^A_t - h^B_t)^2/(2\sigma^2)$.

Claim 10. *If a test has error probabilities* α_0 *under* A *and* β_0 *under* B *with* $\alpha_0 + \beta_0 \le 1$, *then the divergence is at least* $2(1 - \alpha_0 - \beta_0)^2$. *This is Pinsker's inequality applied to the rejection event.*

Claim 11. *Combining Claims 7, 9 and 10:*

$$4\sigma^2(1 - \alpha_0 - \beta_0)^2 \le N\,C_U^2(\Delta\eta)^{4K-2},$$

uniformly in T.

Claim 12. *An estimator that recovers the weights to within half their (fixed,* $\Delta\eta$*-free) difference between the two economies, with probability* $\ge 3/4$ *under each, gives a test with both errors* $\le 1/4$. *So* $N \ge \sigma^2(\Delta\eta)^{-(4K-2)}/C_U^2$.

Precise form. "The weights" means the weight vector listed in increasing order of rate. The two economies' smallest-rate weights differ by a fixed amount that does not shrink with $\Delta\eta$ (for the Lean pair, $2LW/4^L$ with $L = K - 1$). Accuracy is strict: error $<$ half that difference in every coordinate. Is this the right reading of the paper's statement?

Step 6: Wasserstein version.

Claim 13. *The two merged measures are images of the fixed normalized rules under* $s \mapsto r_c + \Delta\eta s$, *scaled to mass* W. *Their* W_1 *distance is* $2c_K W\Delta\eta$ *with* $2c_K = W_1(\mathit{Gauss}, \mathit{Gauss–Radau}) > 0$. *An estimator with* W_1 *error below* $c_K W\Delta\eta$ *(probability* $\ge 3/4$ *under each) gives a test with errors* $\le 1/4$, *and Claim 11 applies.*

3 What Lean has checked, and what it has not

The theorem `p1_thm_kcohort_lower_bound_uniform` in `Paper1KCohortUniform.lean` was built from source with Lean 4 and Mathlib. It uses no `sorry` and only Lean's standard axioms. In mathematical notation it states the following.

> Let $K = L+1 \geq 1$, $c, d \geq 0$ with $0 < c+d < 1$, $W \geq 0$, and $|a|+|\kappa|W/(1-c-d) \leq \varsigma < 1$. Let the two economies have memory rates $c + d\,x_k$ at the $2K$ equally spaced nodes $x_k = 2k/(2K-1) - 1$ of $[-1, 1]$: economy A takes the odd-indexed nodes, economy B the even-indexed ones. Give node k the weight $W\binom{2K-1}{k}/2^{2K-2}$. Let P_A, P_B be the laws of N independent reset experiments of T taps with $N(0, \sigma^2)$ noise, where the responses are computed from the paper's actual state-space model. Then for every measurable event R with $P_A(R) \leq \alpha_0$, $P_B(R^c) \leq \beta_0$ and $\alpha_0 + \beta_0 \leq 1$, and for every T and N,
>
> $$4\sigma^2(1 - \alpha_0 - \beta_0)^2 \leq N\Big(\frac{2\kappa^2 W}{(1-c-d)^{2K}(1-\varsigma)^2}\, d^{2K-1}\Big)^2.$$

Correspondence. Here $c = r_c$ and $d = \Delta\eta$, and $c + d$ is the largest memory rate. The constant equals C_U of Claim 7. In the code the "rate" variable is the gain $\eta = 1-$memory, so the memory is `1 - rate`.

Differences from the paper, for you to judge.

1. **A different pair.** Lean uses the binomial odd/even split of an equally spaced grid, not Gauss/Gauss–Radau. It has matched moments below order $2K - 1$ by the vanishing of the $(2K - 1)$-th forward difference. The theorem asserts existence of *some* pair, so any valid pair proves it. Lean checks the Gauss and Gauss–Radau rules themselves only for $K = 3$.

2. **The estimator statements.** The reductions in Claims 12 and 13 are also checked for this pair (`p1_thm_kcohort_lower_bound_weights`, `p1_thm_kcohort_lower_bound_W1`). The weights are listed in order of memory rate, and the accuracy is $LW/4^L$ per coordinate. The W_1 version uses the event that every 1-Lipschitz function vanishing at zero is integrated to within $2dW/(\pi(2K - 1))$. Please check that this event is the right formalization of "W_1 error below $c_K W \Delta\eta$".

3. **Only the upper order.** The lower order of the response gap (Claim 8) is on paper. It is not needed for the sample bound.

4. **Absolute values in the stability condition** ($|a|$, $|\kappa|$). The paper has $a \geq 0$ and $\kappa > 0$, so the conditions coincide.

5. **The observation model** (Gaussian noise, i.i.d. across taps and experiments, reset from rest) is as in the paper.

Earlier formal versions (`p1_thm_kcohort_lower_bound_every_K_general`) had a constant that grew with T (a factor 4^T). The version above removes that dependence. The paper's earlier proof said only that "analyticity of the kernel-to-response map gives" the response gap. Steps 2–4 above replace that sentence.

4 Questions most likely to expose a gap

1. The constants are uniform only on a spacing range where the stability index ς stays below 1 by a fixed margin. Is that restriction stated clearly enough?

2. The proof uses the hypothesis as an ℓ_1 small-gain bound, $a + \kappa\|k\|_1 < 1$, not only as Schur stability. The paper says the hypothesis makes every economy "Schur stable". Is anything in the proof, or in later uses of the theorem, relying on Schur stability alone, where the small-gain bound might fail?

3. In Claim 12, does "recovering the weights" need an ordering convention to be well defined?

4. Is anything lost by stating the bound for a test rather than for the estimation loss?

5. The case $T = \infty$ is covered as the limit of the uniform-in-T bound: every finite-record experiment obeys it. A literal infinite record would need an infinite Gaussian product experiment and its divergence formula. Is the limiting reading acceptable?

6. The paper says the same order holds for any probe of bounded energy, through $\sup_\omega |\Delta\hat{H}|$. Is that argument complete as written (main text, "What counts as an experiment")?

Response table

Claim	Status	Comment
1–4 (kernel)		
5–7 (response)		
8 (lower order)		
9–11 (testing)		
12–13 (reductions)		
Lean statement vs paper		
Questions 1–6		

Name (optional): _____ Date: _____

Please return to Pratik Niroula. Status options: checked / objection / corrected / open.